Assignment - 5

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Python 3.13.3

GithubLiink:
https://github.com/harshita-0310/AI-ML-Applications-in-Civil-Engineering-CE29208-/tree/main/Assignment-05

Introduction:

Streamflow forecasting plays an important role in water resource planning, flood management, and reservoir operation. Accurate prediction of future streamflow helps in efficient decision-making for irrigation, hydropower generation, and flood control. Artificial Neural Networks (ANN) are widely used in hydrological modeling due to their ability to capture nonlinear relationships between variables. In this study, an ANN model is developed to forecast monthly streamflow using lagged streamflow inputs.

Objective:

To develop an ANN model using lagged streamflow data to predict one-month-ahead streamflow and identify the best lag input combination.

Data Description:

The dataset was structured into six lag configurations ranging from one-month lag to six-month lag inputs. This approach enables evaluation of how past hydrological information influences future streamflow prediction.

The prepared datasets were stored in Excel format and used as input for ANN model development. Each sheet contains cleaned and organized time-series data for model training and testing.

Data Used:

Monthly streamflow dataset from Bhadra reservoir was used.
Lagged datasets (Lag1 to Lag6) were prepared, where:

- Lag1 → previous 1 month streamflow
- Lag2 → previous 2 months streamflow
- Lag3 → previous 3 months streamflow
- Lag4 → previous 4 months streamflow
- Lag5 → previous 5 months streamflow
- Lag6 → previous 6 months streamflow

Each dataset contained input lag variables and target streamflow output.

Methodology:

The ANN model was implemented using NumPy to understand the internal working of neural networks, including forward propagation and backpropagation.The sigmoid activation function was selected due to its ability to model nonlinear relationships and produce bounded outputs.

Training was performed iteratively using gradient-based weight updates. Model performance was monitored using error metrics to ensure convergence.Lag-based modeling was adopted to determine the optimal historical time window required for accurate prediction.

Artificial Neural Network model with one hidden layer was developed using NumPy. Lagged streamflow inputs from one to six months were considered. Data was normalized using Min–Max scaling and divided into training and testing subsets. Model performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2).

The following steps were carried out:

1. Lag datasets were loaded from Excel.
2. Data was cleaned and converted into numeric format.
3. Min–Max normalization was applied to scale the data.
4. Dataset was divided into training (80%) and testing (20%).

5. ANN model was developed using NumPy.
6. Hidden layer with 10 neurons and sigmoid activation function was used.
7. Backpropagation algorithm used for training the network.
8. Model performance evaluated using Mean Squared Error (MSE).
9. ANN models were trained separately for Lag1 to Lag6 datasets.
10. Prediction graphs were generated for each lag.

ANN Working Principle:

Artificial Neural Networks consist of interconnected layers of neurons that process input data and learn relationships through weight adjustments.

The network used in this study includes:

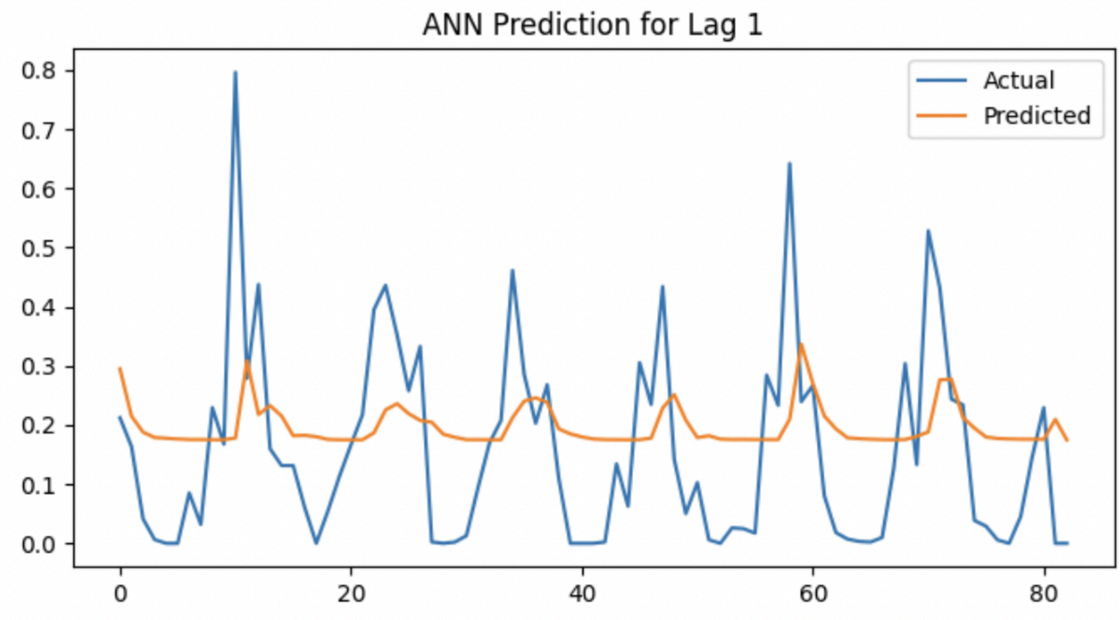
- Input layer receiving lagged streamflow values
- Hidden layer performing nonlinear transformations
- Output layer producing predicted streamflow

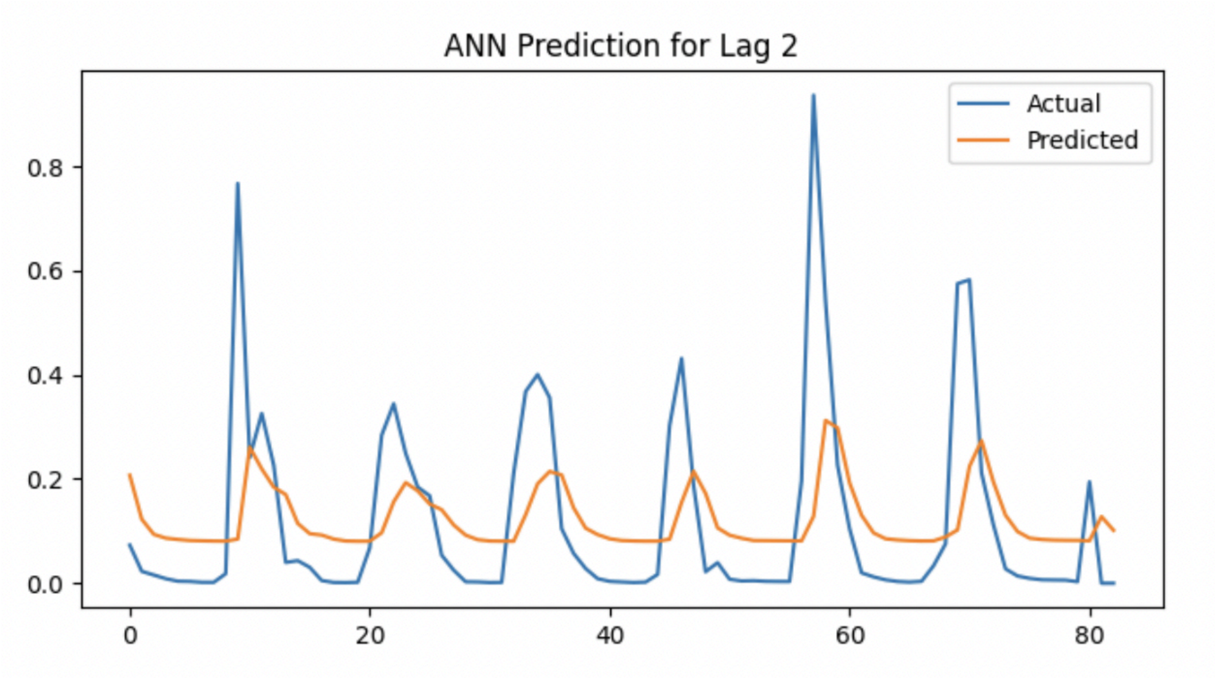
The learning process involves minimizing prediction error through backpropagation.

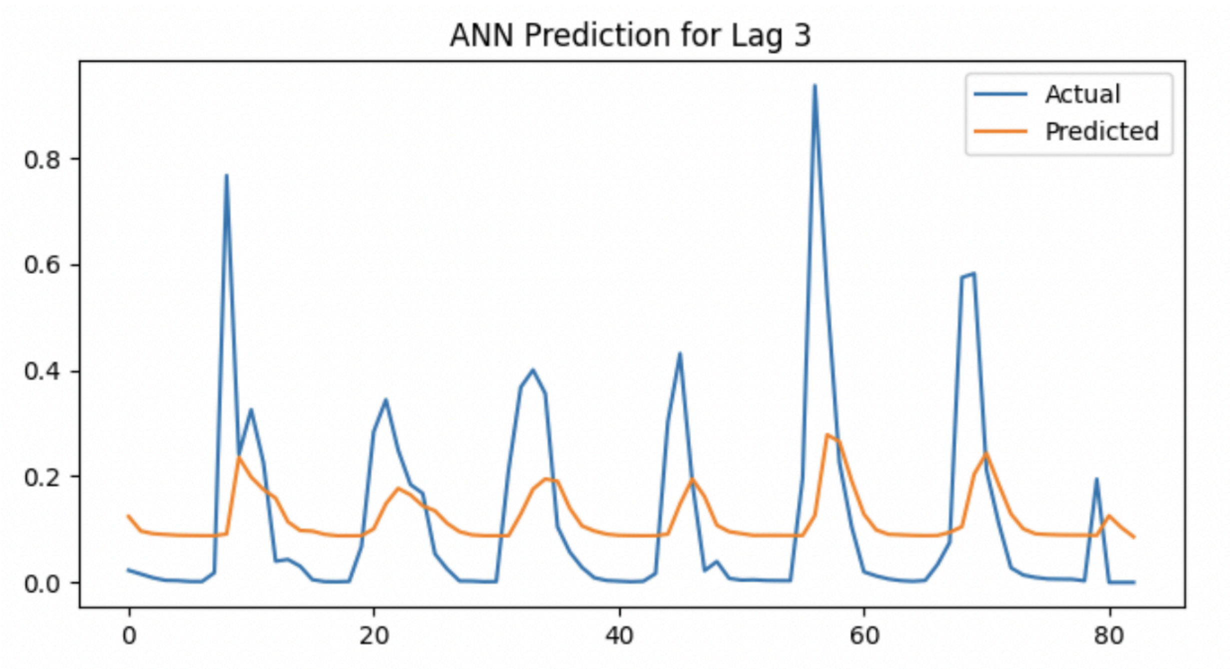
Results and Discussion:

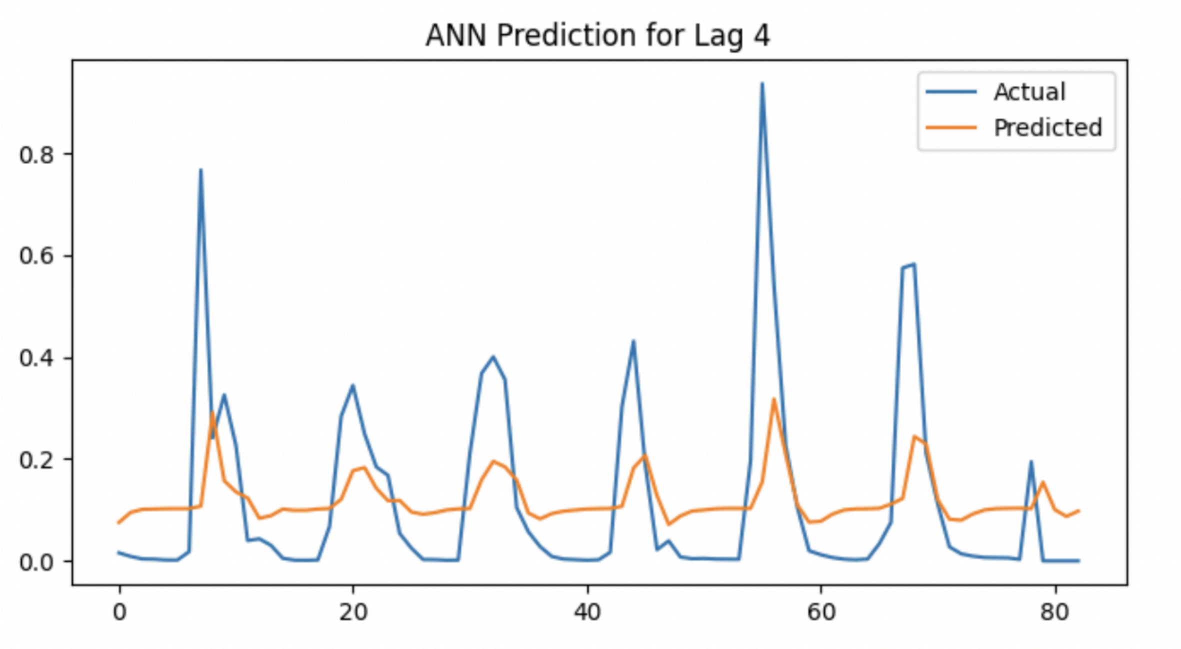
	RMSE	R2
1	0.160099	0.042741
2	0.174903	0.110536
3	0.164723	0.214276
4	0.167451	0.189386
5	4.992570	NaN
6	0.176799	0.098028

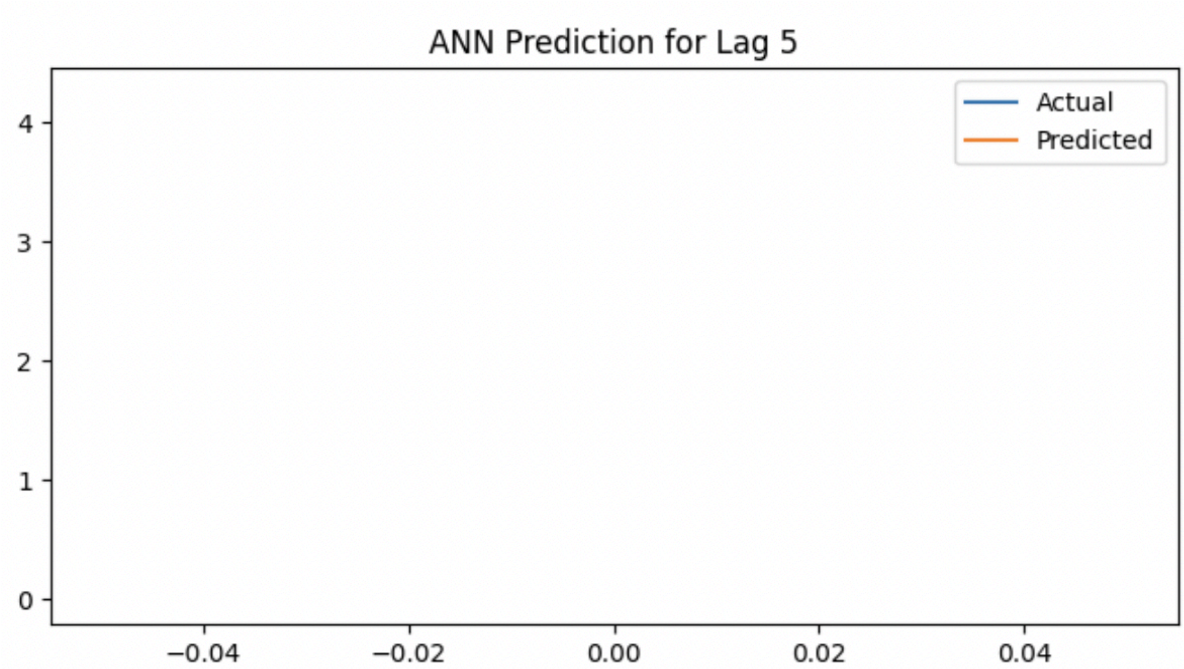
	Lag	MSE
0	1	0.025190
1	2	0.029623
2	3	0.030365
3	4	0.028162
4	5	30.045201
5	6	0.030103

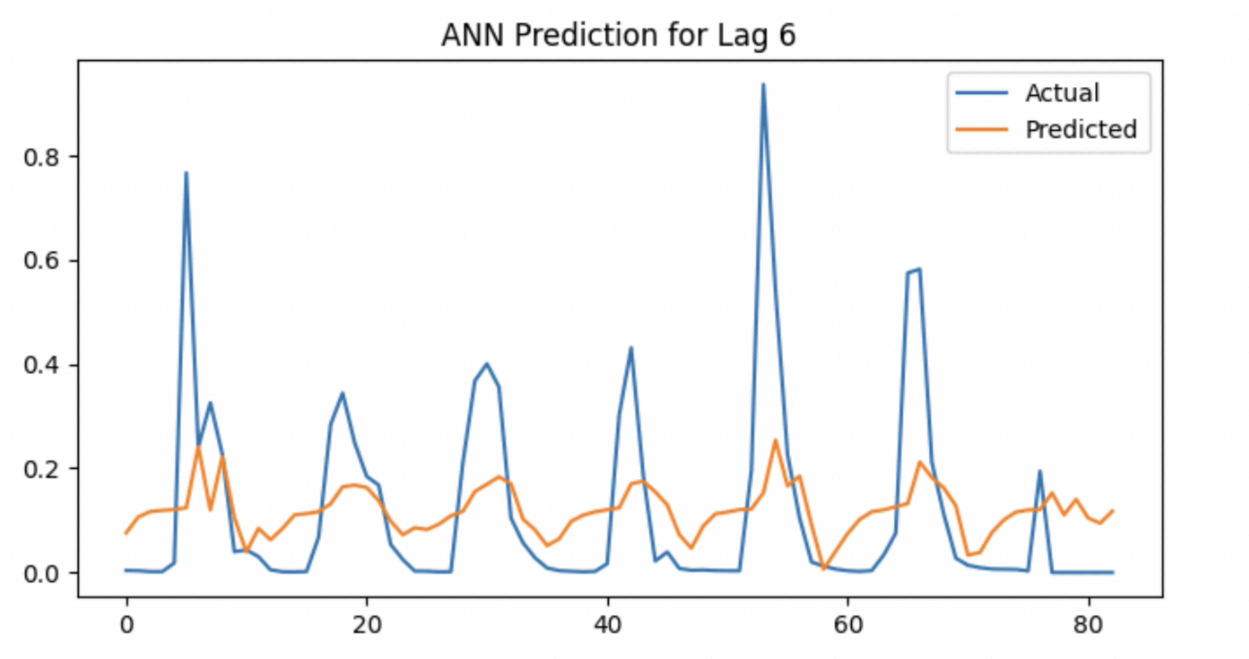



ANN Prediction for Lag 2
Actual
Predicted
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80


ANN Prediction for Lag 3
Actual
Predicted
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80


ANN Prediction for Lag 4
Actual
Predicted
0.0
0.2
0.4
0.6
0.8
0
20
40
60
80


ANN Prediction for Lag 5
Actual
Predicted
0
1
2
3
4
−0.04
−0.02
0.00
0.02
0.04



Model performance was evaluated using RMSE and R^2. Lower RMSE and higher R^2 indicate better prediction accuracy and model fit.

Among all lag combinations, lag = 3 showed the lowest RMSE and highest R^2, indicating the best forecasting performance. This suggests that current streamflow is strongly influenced by the previous three months of data.

Lower lags provided limited historical information, while higher lags introduced noise and reduced prediction accuracy. Lag5 showed comparatively unstable results due to fewer usable samples after preprocessing.

Overall, the ANN model effectively captured the relationship between past and future streamflow, and **lag = 3 was identified as the optimal lag** for prediction.Increasing lag beyond this reduced performance due to noise and reduced effective sample size.The Lag5 dataset showed comparatively unstable results due to limited usable samples after preprocessing and normalization(hence the empty graph)

The ANN model successfully captured the nonlinear relationship between past and future streamflow values.

The prediction plots demonstrate the ability of ANN to capture trends and fluctuations in streamflow. The model predictions closely follow observed values for most lag combinations.

The comparison across lag configurations highlights the importance of selecting appropriate historical input windows. Optimal lag selection improves prediction accuracy and model stability.

Model Interpretation:

The ANN model successfully learned patterns in historical streamflow data and reproduced seasonal and temporal variations.The improvement in prediction accuracy for specific lag combinations indicates that past streamflow values significantly influence current discharge behaviour.

The trained network parameters reflect the nonlinear dependency between past and future hydrological states.

Conclusions:

The ANN model successfully predicted monthly streamflow using lagged inputs.Among all lag combinations, **lag = 3** performed best and provided minimum prediction error.

The study demonstrates that ANN is an effective tool for streamflow forecasting and can assist in water resource planning and reservoir management.